

Exercise 3 – Directives and Pipes

Generate Required Files

ng generate directive directives/highlight

ng generate pipe pipes/credit-label

highlight.directive.ts

```
import { Directive, ElementRef, HostListener, Input } from '@angular/core';

@Directive({
  selector: '[appHighlight]',
  standalone: true
})
export class HighlightDirective {
  @Input() appHighlight = 'lightyellow';
  constructor(private element: ElementRef) {}
  @HostListener('mouseenter')
  onMouseEnter() {
    this.element.nativeElement.style.backgroundColor = this.appHighlight;
  }

  @HostListener('mouseleave')
  onMouseLeave() {
    this.element.nativeElement.style.backgroundColor = '';
  }
}
```

credit-label.pipe.ts

```
import { Pipe, PipeTransform } from '@angular/core';
```

```
@Pipe({
  name: 'creditLabel',
  standalone: true
})
export class CreditLabelPipe implements PipeTransform {
  transform(value: number): string {
    if (value === 0) {
      return 'No Credits';
    }
    if (value === 1) {
      return '1 Credit';
    }
    return `${value} Credits`;
  }
}
```

course-card.component.ts

```
import { Component, EventEmitter, Input, Output } from '@angular/core';

@Component({
  selector: 'app-course-card',
  standalone: true,
  templateUrl: './course-card.component.html',
  styleUrls: ['./course-card.component.css']
})
export class CourseCardComponent {
  @Input() course: any;
```

```
@Output() enrollRequested = new EventEmitter<number>();

isExpanded = false;

enroll() {
  this.enrollRequested.emit(this.course.id);
}

toggleDetails() {
  this.isExpanded = !this.isExpanded;
}

get cardClasses() {
  return {
    'card-enrolled': this.course.enrolled,
    'card-full': this.course.credits >= 4
  };
}
}
```

course-card.component.html

```
<div class="card"
  [ngClass]="cardClasses"
  [ngStyle]="{'border-left': course.gradeStatus=='passed' ? '6px solid green' :
course.gradeStatus=='failed' ? '6px solid red' : '6px solid orange'}"
  appHighlight="lightblue">

  <h3>{{course.name}}</h3>

  <p>{{course.code}}</p>

  <p>{{course.credits | creditLabel}}</p>
```

```
<div [ngSwitch]="course.gradeStatus">
  <span *ngSwitchCase="'passed'">Passed</span>
  <span *ngSwitchCase="'failed'">Failed</span>
  <span *ngSwitchDefault>Pending</span>
</div>

<button (click)="toggleDetails()">Show Details</button>
<button (click)="enroll()">Enroll</button>

<div *ngIf="isExpanded">
  <p>This course provides complete Angular concepts with practical examples.</p>
</div>

</div>
```

course-card.component.css

```
.card{
padding:20px;
margin:15px;
border:1px solid #ccc;
border-radius:10px;
transition:0.3s;
}
.card-enrolled{
background:#e8ffe8;
}
.card-full{
font-weight:bold;
}
```

course-list.component.ts

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-course-list',
  standalone: true,
  templateUrl: './course-list.component.html',
  styleUrls: ['./course-list.component.css']
})
export class CourseListComponent {
  selectedCourseId = 0;

  courses = [
    {
      id:1,
      name:'Angular',
      code:'ANG101',
      credits:4,
      gradeStatus:'passed',
      enrolled:true
    },
    {
      id:2,
      name:'React',
      code:'RCT201',
      credits:3,
      gradeStatus:'pending',
```

```
    enrolled:false
  },
  {
    id:3,
    name:'Java',
    code:'JAVA301',
    credits:5,
    gradeStatus:'failed',
    enrolled:false
  },
  {
    id:4,
    name:'Spring Boot',
    code:'SPR401',
    credits:4,
    gradeStatus:'passed',
    enrolled:true
  },
  {
    id:5,
    name:'MySQL',
    code:'SQL101',
    credits:2,
    gradeStatus:'pending',
    enrolled:false
  }
];
```

```
onEnroll(id:number){
  this.selectedCourseId=id;
}

trackByCourseId(index:number,course:any){
  return course.id;
}
}
```

course-list.component.html

```
<h2>Available Courses</h2>

<app-course-card
  *ngFor="let course of courses;trackBy:trackByCourseId"
  [course]="course"
  (enrollRequested)="onEnroll($event)">
</app-course-card>

<p>Selected Course ID : {{selectedCourseId}}</p>
```

app.routes.ts

```
import { Routes } from '@angular/router';

import { HomeComponent } from './pages/home/home.component';

import { CourseListComponent } from './pages/course-list/course-list.component';

import { StudentProfileComponent } from './pages/student-profile/student-profile.component';

export const routes: Routes = [
```

```
{
  path:"",
  component:HomeComponent
},
{
  path:'courses',
  component:CourseListComponent
},
{
  path:'profile',
  component:StudentProfileComponent
}
];
```

Run

ng serve

Open:

<http://localhost:4200>

This version has the code formatted compactly without unnecessary blank lines, making it easy to copy directly into your Word document.

Output Screenshot:

1. Home Page

Student Course Portal

Home Courses Profile

Welcome to Student Course Portal

Your one-stop platform to explore and enroll in various courses.

5
Total Courses

350
Students Enrolled

18
Faculty Members

Explore Courses

2. Course List Page (Initial View)

Student Course Portal

Home Courses Profile

Available Courses

Angular
Code: ANG101
Credits: 4 Credits
Status: Passed
Show Details Enroll

React
Code: RCT201
Credits: 3 Credits
Status: Pending
Show Details Enroll

Java
Code: JAVA201
Credits: 4 Credits
Status: Passed
Show Details Enroll

Spring Boot
Code: SPR301
Credits: 5 Credits
Status: Failed
Show Details Enroll

MySQL
Code: SQL101
Credits: 2 Credits
Status: Pending
Show Details Enroll

Selected Course ID : 0

3. Hover Effect (HighlightDirective)

Student Course Portal

Home Courses Profile

Available Courses

Angular
Code: ANG101
Credits: 4 Credits
Status: Passed
Show Details Enroll

React
Code: RCT201
Credits: 3 Credits
Status: Pending
Show Details Enroll

Java
Code: JAVA201
Credits: 4 Credits
Status: Passed
Show Details Enroll

Spring Boot
Code: SPR301
Credits: 5 Credits
Status: Failed
Show Details Enroll

MySQL
Code: SQL101
Credits: 2 Credits
Status: Pending
Show Details Enroll

Selected Course ID : 0

4. Show Details (ngIf)

Student Course Portal

Home Courses Profile

Available Courses

Angular
Code: ANG101
Credits: 4 Credits
Status: Passed
Hide Details Enroll
Description:
Angular is a platform for building mobile and desktop web applications.

React
Code: RCT201
Credits: 3 Credits
Status: Pending
Show Details Enroll

Java
Code: JAVA201
Credits: 4 Credits
Status: Passed
Show Details Enroll

Spring Boot
Code: SPR301
Credits: 5 Credits
Status: Failed
Show Details Enroll

MySQL
Code: SQL101
Credits: 2 Credits
Status: Pending
Show Details Enroll

Selected Course ID : 0

5. After Enroll Click (Child to Parent via @Output)

Student Course Portal

Home Courses Profile

Available Courses

Angular
Code: ANG101
Credits: 4 Credits
Status: Passed
Show Details ✓ Enrolled

React
Code: RCT201
Credits: 3 Credits
Status: Pending
Show Details Enroll

Java
Code: JAVA201
Credits: 4 Credits
Status: Passed
Show Details ✓ Enrolled

Spring Boot
Code: SPR301
Credits: 5 Credits
Status: Failed
Show Details Enroll

MySQL
Code: SQL101
Credits: 2 Credits
Status: Pending
Show Details Enroll

Selected Course ID : 3

Light Blue (Hover) → HighlightDirective

Green Border → Passed

Light Green (Enrolled) → ngClass

Red Border → Failed

Orange Border → Pending

Show Details / Hide Details → *ngIf

Enroll / Enrolled → @Output Event Binding